2022-DSE
BIO
PAPER 1B

B

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Candidate Number									

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2022

BIOLOGY PAPER 1

SECTION B : Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.

(2) Refer to the general instructions on the cover of the Question Paper for Section A.

(3) Answer **ALL** questions.

(4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

(5) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.

(6) Present your answers in paragraphs wherever appropriate.

(7) The diagrams in this section are **NOT** necessarily drawn to scale.

(8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

* A 1 3 0 E 0 1 B *

SECTION B

Answer **ALL** questions. Write your answers in the spaces provided.

1. The following are some life processes in humans:

   Egestion (A)
   Excretion (B)
   Feeding (C)
   Growth (D)
   Respiration (E)

   (a) Use the letters to construct an equation showing the relationship of these processes in energy flow. (1 mark)

   $\boxed{\quad} = \boxed{\quad} - \boxed{\quad} - \boxed{\quad} - \boxed{\quad}$

   (b) In terms of the energy value of each process, explain why the D/C ratio for vegetarians is lower than that for non-vegetarians. (3 marks)

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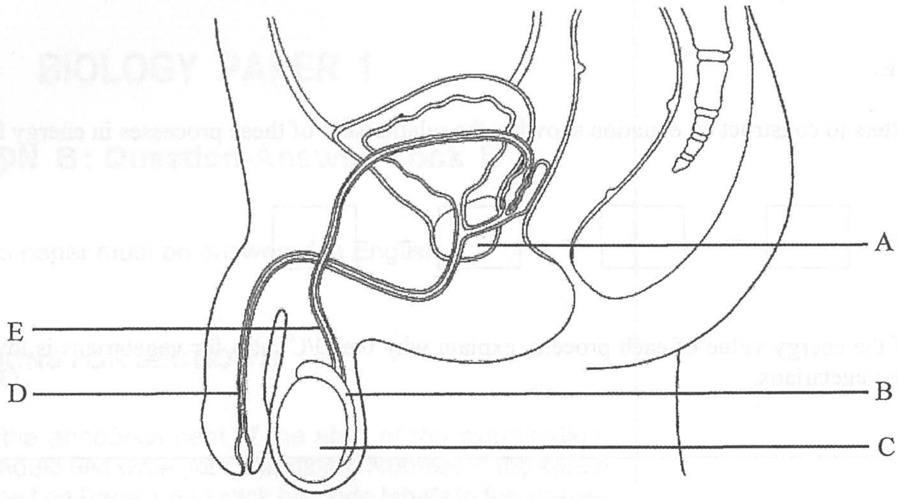
2. The diagram below shows the human male reproductive system and its associated structures:



(a) In which structure does meiosis take place? Give your answer using the letters in the diagram. (1 mark)

(b) State the functions of structures A and B respectively. (2 marks)

(c) Vasectomy is an operation for achieving permanent contraception in males.

(i) Using the letters in the diagram, state the structure which is affected in this contraceptive method. (1 mark)

(ii) What is the biological basis of this contraceptive method? (2 marks)

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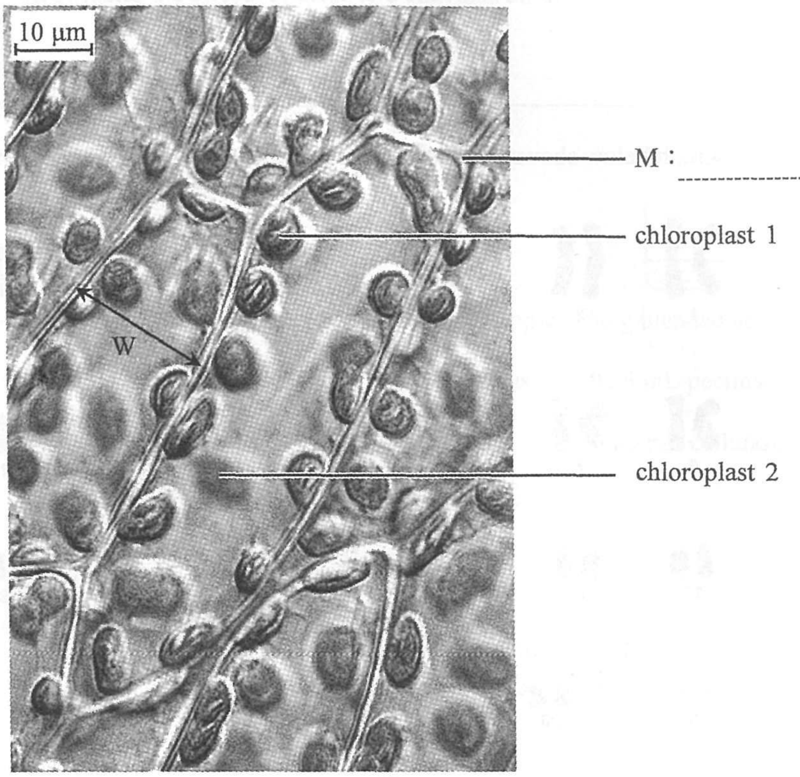
3. The photomicrograph below shows some unstained plant cells:



(a) Label structure M. (1 mark)

(b) What is the actual length of W shown in the photomicrograph? (1 mark)

(c) Chloroplast 1 appears sharp in this photomicrograph while chloroplast 2 appears blurred. To obtain a sharper image of chloroplast 2, how should you operate the microscope? (1 mark)

(d) Some structures of chloroplast cannot be distinguished in this photomicrograph. State *one* of these structures. (1 mark)

(e) Give *one* equipment that can be used to observe the structure stated in (d). (1 mark)

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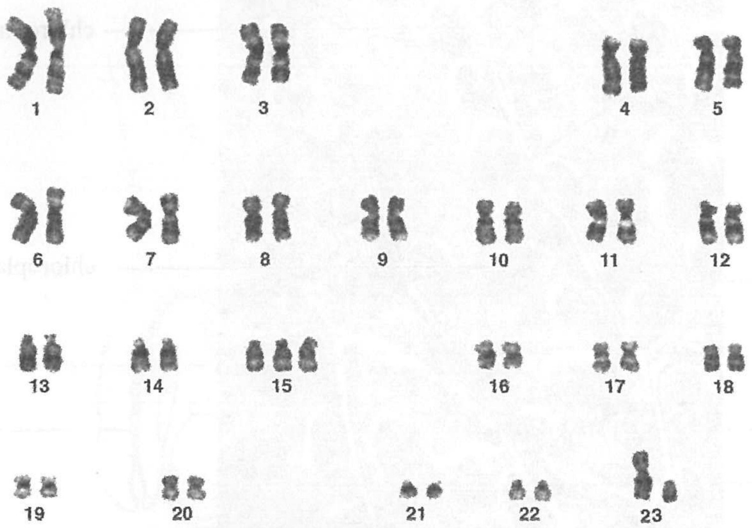
4. The photomicrograph below shows the karyotype of a patient who is suffering from a certain brain disease:



(a) What is the gender of this patient? Describe ***one*** observable feature from the karyotype to support your answer. (2 marks)

(b) (i) Describe the abnormality shown in the karyotype. (1 mark)

(ii) State the type of mutation involved in this abnormality. (1 mark)

(iii) How would this abnormality affect the mRNA level in the brain cells of this patient? (1 mark)

(c) The cerebellum is one of the regions affected by this disease. In relation to the function of the cerebellum, suggest ***one*** difficulty that would be experienced by this patient. (1 mark)

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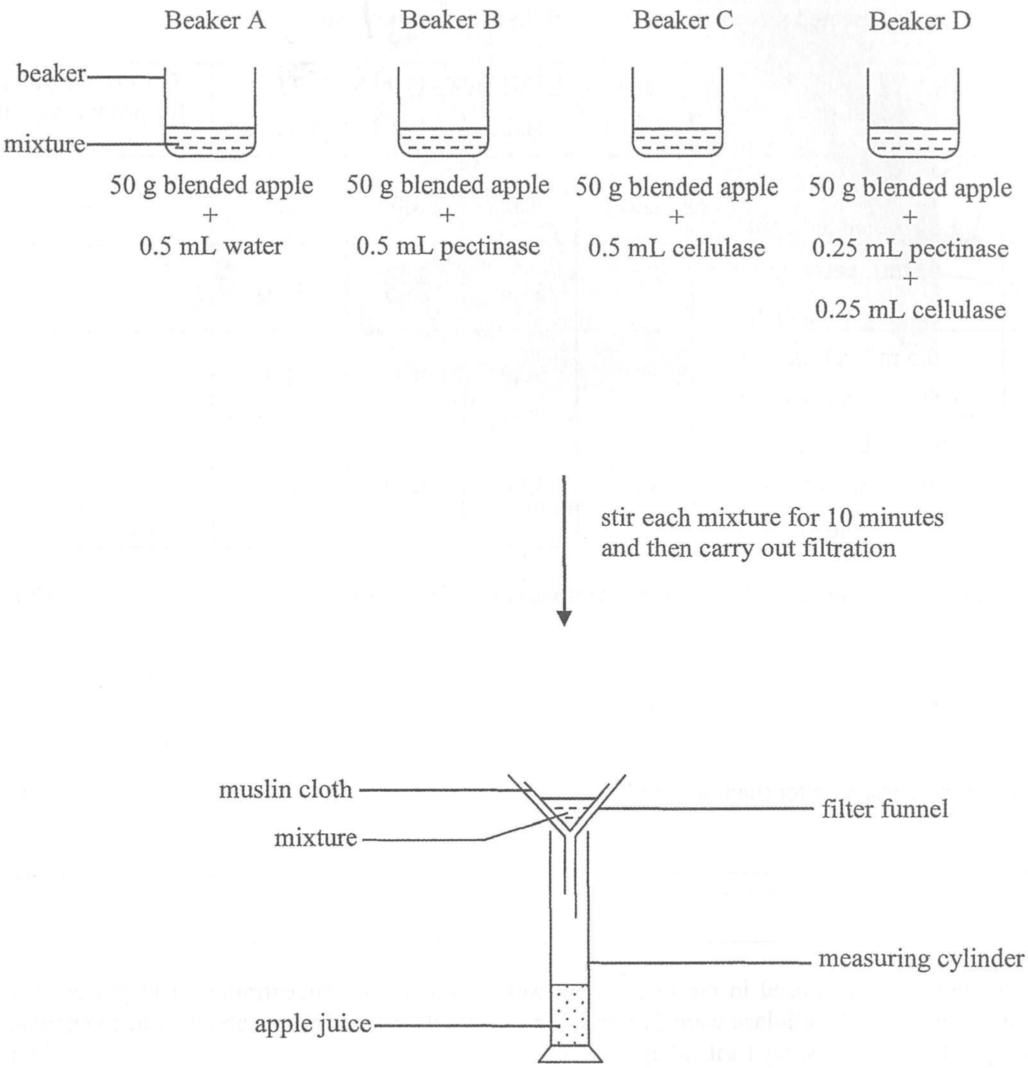
5. Pectinase and cellulase are enzymes that break down the chemical components of plant cell walls. The following experiment investigates the effects of these two enzymes on the production of apple juice:



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Answers written in the margins will not be marked.

The experiment was repeated three times and the results are shown below:

Beaker	Mixture	Volume of apple juice produced (mL)				Cost of enzyme(s) for producing 1 mL apple juice
		Trial 1	Trial 2	Trial 3	Average	
A	0.5 mL water + 50 g blended apple	2.0	1.0	3.0	2.0	---
B	0.5 mL pectinase + 50 g blended apple	33.5	31.0	28.5	31.0	
C	0.5 mL cellulase + 50 g blended apple	4.5	4.0	3.5	4.0	
D	0.25 mL pectinase + 0.25 mL cellulase + 50 g blended apple	34.0	32.0	36.0	34.0	

(a) State the independent variable and dependent variable of this experiment. (2 marks)

(b) Why are three trials better than one trial? (1 mark)

(c) The enzyme solutions used in the experiment were at the same concentration. The prices of 0.5 mL pectinase and 0.5 mL cellulase were \$13 and \$7 respectively. Complete the above table to show the cost of enzyme(s) for producing 1 mL of apple juice. (2 marks)

(d) Based on the answer in (c), which is the most cost-effective means for producing apple juice? (1 mark)

(e) Explain why the apple juice collected from Beaker D is clearer than that from Beaker A. (1 mark)

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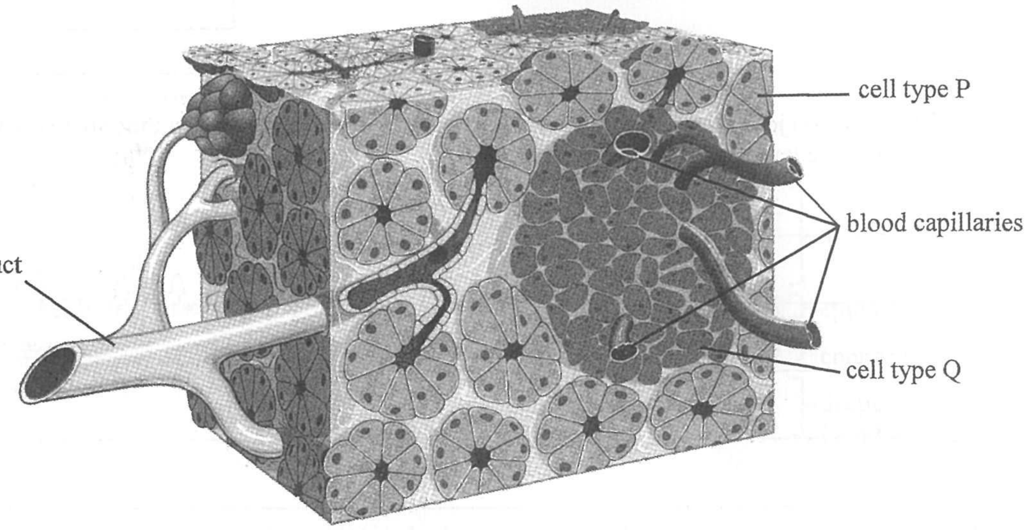
6. The schematic diagram below illustrates the distribution of different cell types in the human pancreas:



(a) Which type of cells, P or Q, secretes hormones? Support your answer with ***one*** observable feature illustrated in the diagram. (3 marks)

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(b) A person jogs slowly for an hour. Describe how the hormones from the pancreas can regulate the blood glucose level of the person while jogging. (4 marks)

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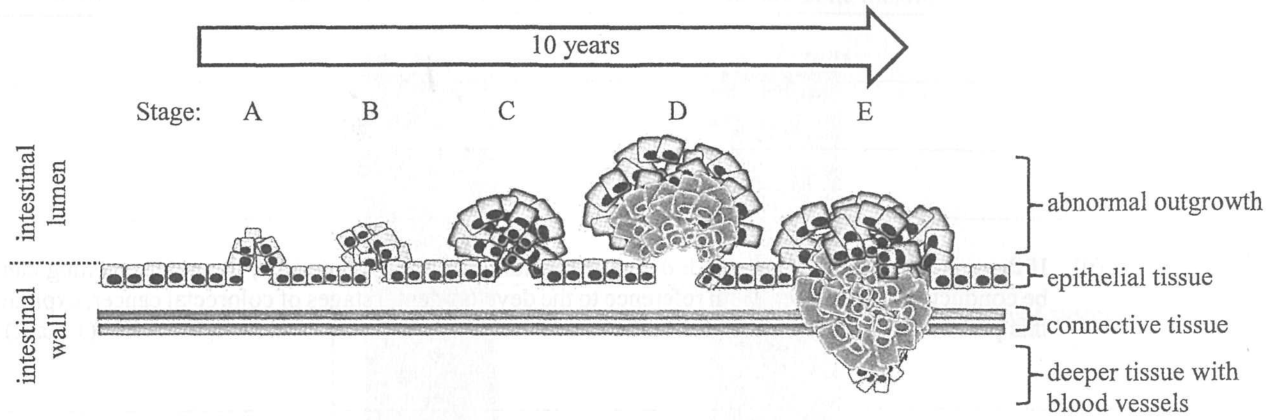
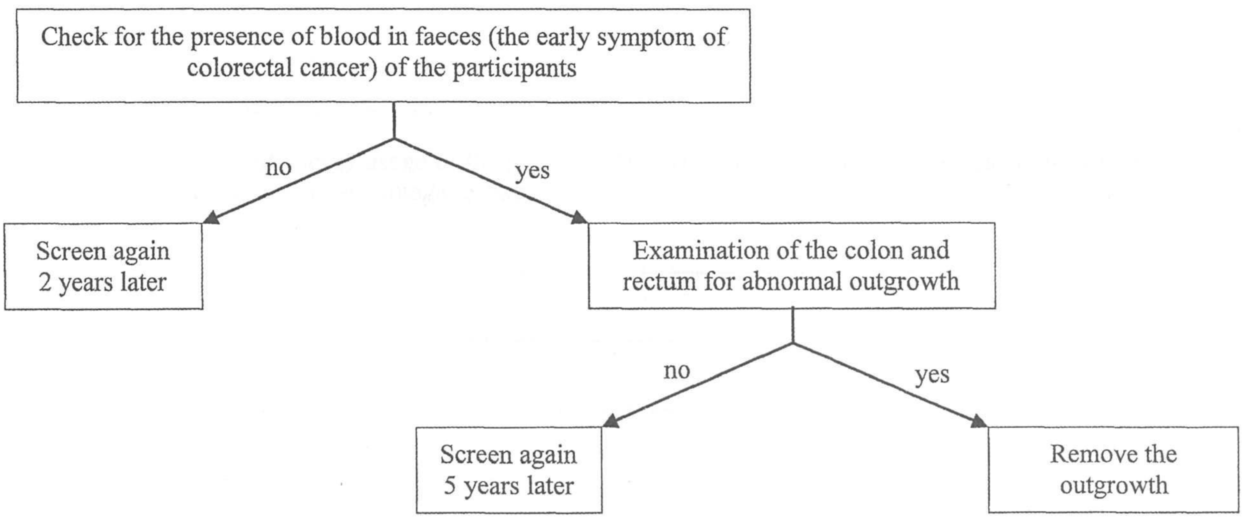
7. Colorectal cancer is one of the most common cancers in Hong Kong. The schematic diagram below shows the developmental stages of colorectal cancer over time:



(a) Which stage of colorectal cancer has a high risk of spreading? Explain your answer. (2 marks)

(b) The Department of Health in Hong Kong has launched a regular screening programme for the prevention of colorectal cancer. The flowchart below illustrates the screening programme:



(i) The screening programme is recommended for people aged 50 or above. Give ***two*** reasons why this group of people is more susceptible to colorectal cancer. (2 marks)

(ii) If there is no abnormal outgrowth of the epithelium of the large intestine, the next screening can be conducted 5 years later. With reference to the developmental stages of colorectal cancer, explain this practice. (1 mark)

(iii) Recently, there is a growing trend of people diagnosed with colorectal cancer at a younger age. Suggest ***two*** eating habits which may lead to this growing trend. (2 marks)

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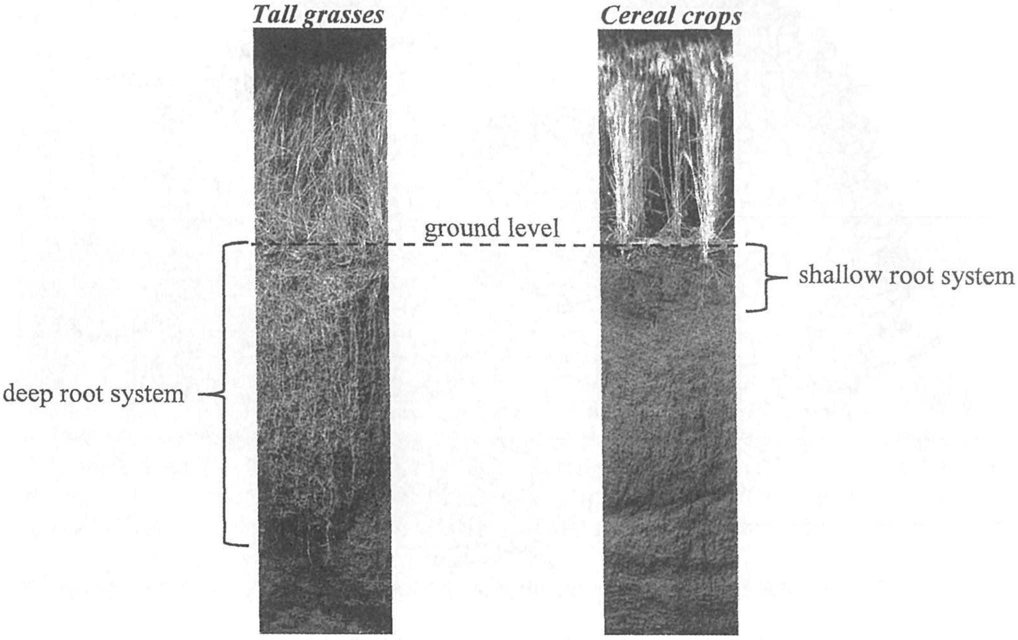
8. Tall grasses and cereal crops belong to the same family and share a common ancestor. Cereal crops have been artificially selected for agriculture and their seeds harvested as food. The roots of the tall grasses in grasslands range in depth from 1.5 m to 4.5 m while those of cereal crops rarely exceed 1 m. The photographs below show the root depths of tall grasses and cereal crops under the same magnification:



(a) (i) Grassland is a region of treeless plain which is mainly occupied by tall grasses. It is usually found in regions with moderate rainfall. The tall grasses have evolved with a deep root system. What is the selection pressure involved in this evolutionary process? What is the advantage of the greater root depth in tall grasses? (2 marks)

(ii) In terms of energy usage of the plant, explain why having a shallow root system for cereal crops is considered an advantage to farmers. (2 marks)

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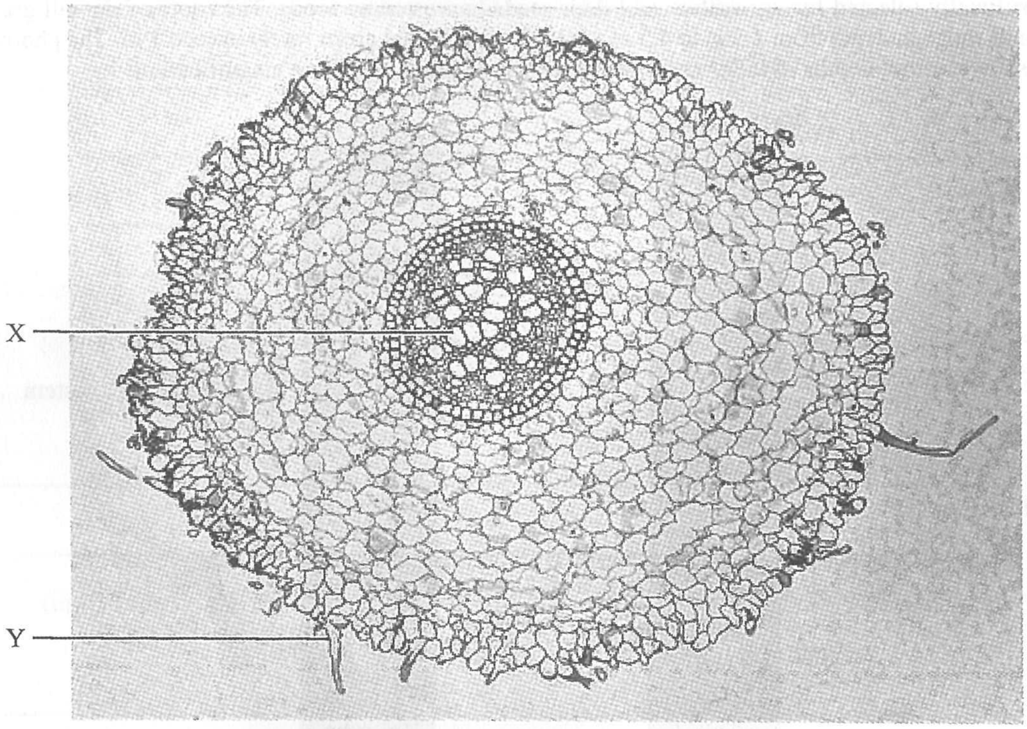
(b) The photomicrograph below shows the cross-section of the root of a cereal crop:



(i) Complete the table below to show how an observable feature of the structures X and Y is related to its function. (4 marks)

	Observable feature	Function
X		
Y		

(ii) Explain how water is transported from structure Y to structure X. (3 marks)

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9. Antibiotics have been widely used to treat bacterial infections. They work by killing bacteria or inhibiting their growth. However, some strains of bacteria have developed resistance to antibiotics.

(a) How can antibiotics kill bacteria or inhibit their growth? State ***three*** possible ways. (3 marks)

(b) Bacterium R is a pathogenic bacterium which possesses an antibiotic resistance gene. Its gene product can break down antibiotic X. Scientists have suggested a new approach to fight against bacterium R. This new approach involves the use of a synthetic polynucleotide which binds to the mRNA transcribed from the antibiotic resistance gene. The expression of the gene is then inhibited. By administering this synthetic polynucleotide together with antibiotic X, bacterium R can be killed.

The base sequences of the synthetic polynucleotide and part of the mRNA are shown below:

synthetic polynucleotide: AGT GAC TCG GTC AGC

mRNA: ... AUG UCU GUU CCA UCA UCA CUG AGC CAG UCG GCC AUU AAU GCC AAC UAG ...

(i) On the mRNA, underline the base sequence to which the synthetic polynucleotide will bind. (1 mark)

(ii) Explain how the synthetic polynucleotide can inhibit the expression of the antibiotic resistance gene. (3 marks)

(iii) Suggest ***one*** advantage of using synthetic polynucleotides to fight against antibiotic resistant bacteria. (1 mark)

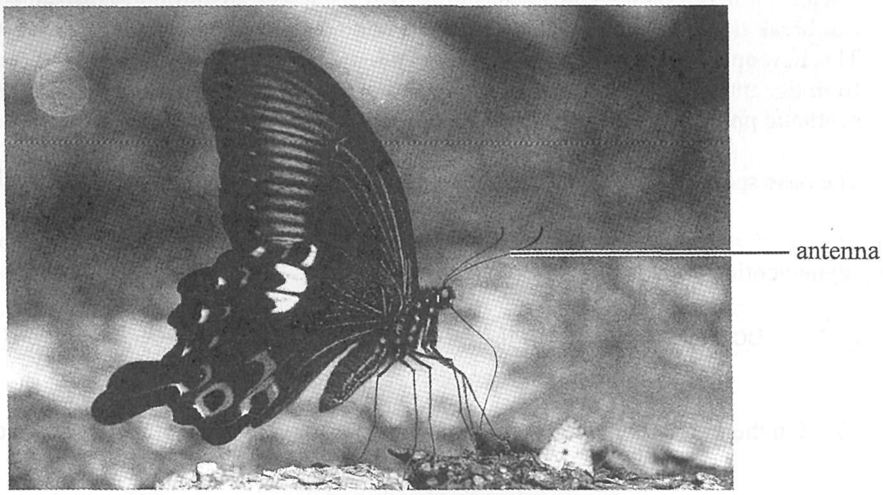
Answers written in the margins will not be marked.

10. *Habenaria* is a genus of orchids with dull-coloured and scented flowers, which attract moths for pollination at night. In the group, *H. rhodocheila* is an exceptional species with reddish flowers which lack a detectable scent. It was found that no insects visited the flowers of *H. rhodocheila* at night for pollination while insect species A consumed nectar from the flowers in the daytime.

(a) The following dichotomous key can be used to identify the group to which insect species A belongs:

1a	with wings	....	2
1b	without wings	....	Group P
2a	antennae longer than head	....	3
2b	antennae shorter than head	....	Group Q
3a	wings rest at their sides	....	Group R
3b	wings rest together upright	....	Group S

The photograph below shows the appearance of species A at rest. Using the above key, write down the sequence which leads to the identification. (1 mark)

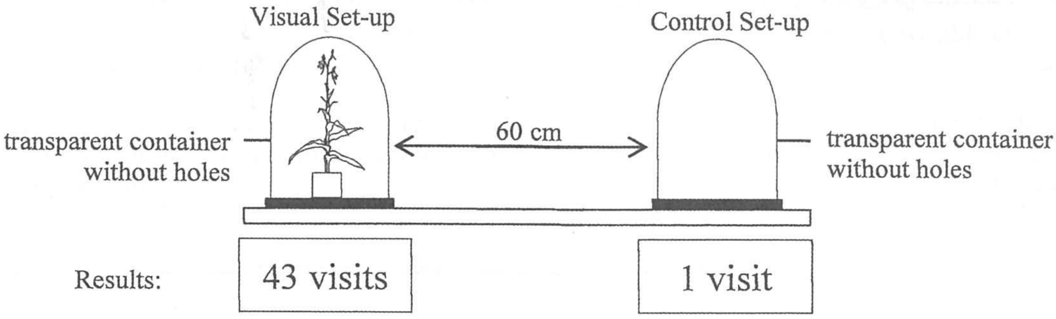


(b) In this case, *H. rhodocheila* was evolved from other *Habenaria* species. Suggest how the speciation of *H. rhodocheila* could be facilitated by different insect pollinators. (3 marks)

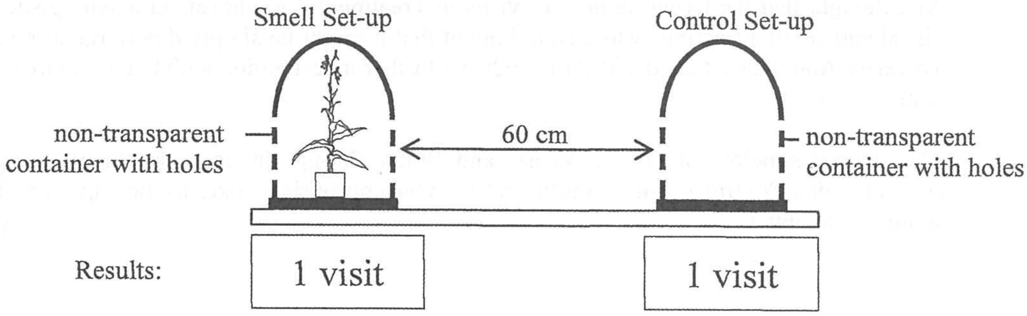
Answers written in the margins will not be marked.

(c) To determine if insect species A is attracted to the flowers of *H. rhodocheila* by its appearance or its smell (if any), researchers designed an investigation as shown in the diagrams below:

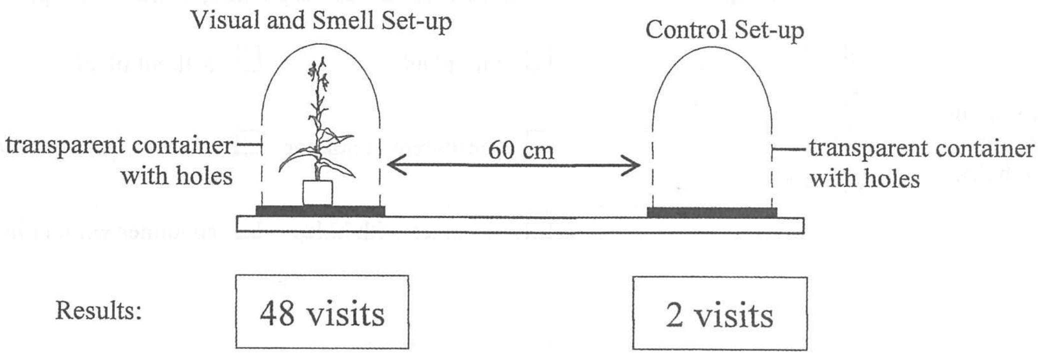
Treatment 1:



Treatment 2:



Treatment 3:



In each treatment, two set-ups were placed in an open area at a distance of 60 cm from each other for one hour. After that, the position of the two set-ups was exchanged for another hour. During the two-hour period, if an individual of species A approached any set-up to a distance of less than 10 cm, it was counted as one visit. The number of visits to each set-up is shown in the boxes in the above diagrams.

(i) To ensure that the investigation was a fair test, exchanging the positions of the two set-ups was a necessary step. Explain why. (1 mark)

....

....

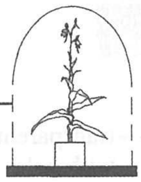
(ii) With reference to the aim of the investigation, what conclusions can you draw from the results of Treatment 1 and Treatment 2 respectively? (4 marks)

(iii) When Ann and Ken compared the results of the three treatments, they had different interpretations. Ann thought that the larger number of visits in Treatment 3 might reflect a synergistic effect of visual and smell attractions while Ken thought that it might be simply due to random variations. To verify Ann's idea, they decided to conduct a further investigation with two more treatments for comparison.

The diagrams below show the Visual and Smell Set-up. In each treatment, complete the corresponding Control Set-up by putting a '✓' in the appropriate boxes to show the conditions that should be adopted. (2 marks)

Treatment 4:

Visual and Smell Set-up



Conditions for the corresponding Control Set-up:

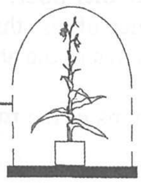
☐ with plant ☐ without plant

☐ transparent container ☐ non-transparent container

☐ container with holes ☐ container without holes

Treatment 5:

Visual and Smell Set-up



Conditions for the corresponding Control Set-up:

☐ with plant ☐ without plant

☐ transparent container ☐ non-transparent container

☐ container with holes ☐ container without holes

Answers written in the margins will not be marked.

You are required to present your answer to the following question in essay form. Criteria for marking will include relevant content, logical presentation and clarity of expression.

11. Recently, the use of the ketogenic diet for achieving weight loss is becoming popular. In fact, this high-fat, moderate-protein and very-low-carbohydrate diet has been used as an approach to control the blood glucose level in diabetics. However, the effectiveness of this diet in achieving weight loss is still controversial.

Describe how a ketogenic diet can be used to control the blood glucose level in diabetics. Evaluate the possibility of using this diet for weight loss and discuss the health concerns of adopting such a diet for healthy persons. (12 marks)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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